

Integrating Vision–Language–Action Control into Forklift Unloading through Data-Efficient Adaptation via Simulation

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Abstract—Forklifts are widely used in the warehousing and distribution sector and play a critical role in modern global supply chains. Despite advances in forklift automation, existing systems remain limited in handling complex tasks that require sequential decision-making. Recently, vision–language–action (VLA) models have emerged as a promising approach for directly generating control actions from visual observations and language prompts, although their application to forklifts remains unexplored. Therefore, this study demonstrates a data-efficient integration of VLA control for an autonomous forklift unloading task. Moreover, training VLA models typically requires large-scale datasets. As there is currently no public dataset for forklift operations, we use a VLA model pretrained primarily on robotic manipulator data and construct a training dataset via simulation. We then adapt the pretrained VLA model to the distinct embodiment of a forklift using this data-efficient training approach, in which real-world demonstrations account for approximately 8% of the data. The integrated system was validated through 72 real-world trials across four model variants. The results show that the proposed system can accomplish an unloading task with limited real-world data. This study demonstrates an initial real-world integration of VLA-based forklift automation.

I. INTRODUCTION

Forklifts are among the most widely used industrial vehicles in warehousing and distribution, and their automation has become increasingly important in the context of rapidly expanding global logistics. Forklift automation has already been implemented in restricted environments using system designs similar to those reported by Walter et al. [1] and Teller et al. [2]. However, these approaches rely heavily on accurate state estimation using multiple sensors and hard-coded logic tailored to constrained spaces [3]. In real-world deployments, where forklifts operate under changing conditions, hard-coding decisions for all possible situations is prohibitively costly to engineer and maintain. This motivates the need for an alternative approach to conventional systems that depend heavily on human-engineered design.

Recently, vision–language–action (VLA) models have emerged as a promising approach for decision-making tasks that are difficult to handle with conventional methods [4], [5], [6], [7]. This capability has enabled automation in long-horizon and complex tasks [8], [9], [10], [11], and may also enable automation in complex forklift operations while reducing reliance on human-engineered design. However, forklift-specific studies of VLA-based control remain limited.

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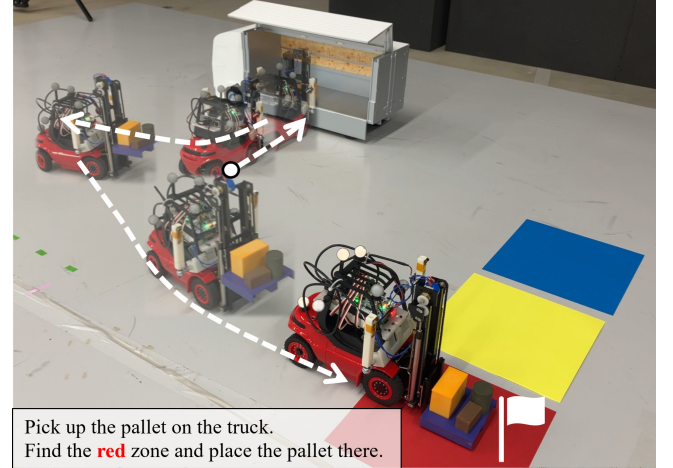


Fig. 1: Behavior of the 1/14-scale forklift executing an unloading task using our VLA model. Given only a language prompt, the forklift autonomously retrieves a pallet from the wing truck, reverses while searching the right side, and places the pallet at the location corresponding to the color specified in the prompt.

In this study, we integrate a VLA model into a forklift unloading system to automate the unloading sequence by directly generating actions from language and visual observations (Fig. 1). Our integrated system can handle task goals in natural language and generate action sequences while jointly predicting the current task phase, eliminating the need to manually enter numeric targets or implement extensive hard-coded logic. Moreover, training VLA models requires large-scale datasets. However, to the best of our knowledge, no public dataset currently exists for forklift operations. Therefore, we employed a VLA model pretrained primarily on robotic manipulator data and constructed a simulation dataset to bridge the embodiment gap between manipulators and forklifts, thereby facilitating cross-embodiment transfer to forklift operation. We performed a real-world evaluation of our approach on a 1/14-scale forklift platform by conducting 72 trials, including baseline comparisons. Although full-scale forklifts are the intended target, the 1/14-scale platform enables safer, real-world iterations during early-stage learning-based development. To the best of our knowledge, this is the first real-world integration study of VLA-based end-to-end action generation for a forklift unloading task. The main contributions are as follows:

- We integrate a VLA model into a forklift unloading

system, enabling phase inference and action generation from onboard sensor observations while supporting low-latency execution.

- We present a data-efficient cross-embodiment adaptation recipe for transferring a VLA model pretrained on robotic manipulator data to forklift control with a small amount of real-world data.
- We evaluate the proposed integrated system through 72 real-world trials across four model variants and demonstrate that it can accomplish the unloading task with limited real-world data.

II. RELATED WORK

Recent forklift studies have combined advanced sensing technologies with deep learning to improve the robustness of environmental perception and partially reduce reliance on hand-coded logic [12], [13], [14]. However, the transition from perception to task execution still requires manual system design. Reinforcement-learning (RL)-based action generation has also been studied for forklift tasks [3], [15]. Using RL for long-horizon forklift tasks, such as unloading, is difficult owing to reward-design complexity. To address these limitations, we introduce recent VLA techniques to forklift control. VLA models are typically pretrained on large-scale web and robot datasets, then fine-tuned on target-robot data [7]. As existing robot datasets are biased toward robot arms [5], [9] and autonomous driving [16], applying VLA models to industrial machinery, such as forklifts, is challenging. To bridge this gap, we adapt a VLA model for forklift action generation using digital-twin-generated and limited real-world data. Recent VLA studies have also explored joint learning of semantic states with action generation [17], [11] and asynchronous action chunking for real-time deployment [10], [18]. We implement simplified versions of these ideas on a physical forklift platform.

III. SYSTEM OVERVIEW

The proposed integration combines an online VLA-based forklift control system with an offline data-efficient adaptation pipeline, as shown in Fig. 2. As a target task, we address pallet unloading from a wing truck using a 1/14-scale forklift platform for safe and repeatable evaluation.

A. Task setting

The unloading task consists of subtasks, including inserting the forks into the pallet, lifting the pallet, backing out, searching the right side for the target color while backing out, and placing the pallet at the specified color location. We use three color targets (red, yellow, and blue), and the forklift starts approximately 0.15 m in front of the pallet. The operator only provides the prompt; all other operations are executed autonomously.

B. Hardware platform

We used the 1/14-scale forklift platform introduced by Oishi et al. [15]. Most hardware and sensing specifications follow those reported by Oishi et al.; therefore, we describe

only the components relevant to this study. This platform is a counterbalance-type forklift with rear-wheel steering and front-wheel drive. Visual data were captured using two pinhole cameras mounted on the left and right sides of the forklift. The vehicle speed and yaw rate were obtained using a motion-capture system, and the fork height was measured using a rotary encoder mounted near the top of the fork assembly. In a full-scale forklift, these speeds are typically available directly from the vehicle. For actions, the throttle command, steering command (servo motor angle), and fork-height command (target height) were used. The forklift was equipped with a Raspberry Pi 4, which communicates with the inference PC via the Robot Operating System (ROS) [19] to exchange action commands and sensor observations.

C. Integrated architecture

Fig. 2 shows the overall architecture of the proposed system. The offline pipeline first trains a VLA policy on web-scale vision–language and large-scale robot manipulation data and then adapts it to forklift control. Simulator-generated forklift data adapts the policy to the target task, while a small amount of real-world forklift data reduces the remaining environmental and embodiment gaps. This approach reduces the real-world forklift data required for training.

The use of robot manipulation data is also motivated by its structural similarity to forklift operation. Our pallet unloading task is essentially a pick-and-place problem: the forklift lifts a pallet and places it at a designated location. For observation data acquisition, we used cameras mounted on the left and right sides of the forklift, similar to the wrist-mounted cameras on robot arms.

In online operation, the adapted policy receives camera images, vehicle-state observations, and a language prompt, and outputs an action chunk over a future horizon. Let t_{inf} denote the maximum time required for one VLA inference. Because VLA inference is computationally expensive, the system runs inference every t_{inf} and executes forklift actions asynchronously. The policy is trained to predict actions up to $2t_{\text{inf}}$ after the observation time, so that the controller can skip the portion corresponding to inference latency and execute the future part of the predicted action chunk. This design reduces idle waiting time and maintains temporal consistency between observation, inference, and vehicle motion.

IV. MODEL AND TRAINING STRATEGY

We construct the forklift policy based on the adaptation pipeline described in Section III-C. Recent progress in robot manipulation has led to several publicly available VLA models trained on large-scale web and manipulation data. In this work, we use $\pi_{0.5}$ [17] as the base model, which allows us to bypass the initial construction of a VLA policy and focus on adapting it to forklift control.

A. Model architecture

Our model is based on $\pi_{0.5}$ [17] and is reconfigured for forklift operation. We denote the observation at time t as

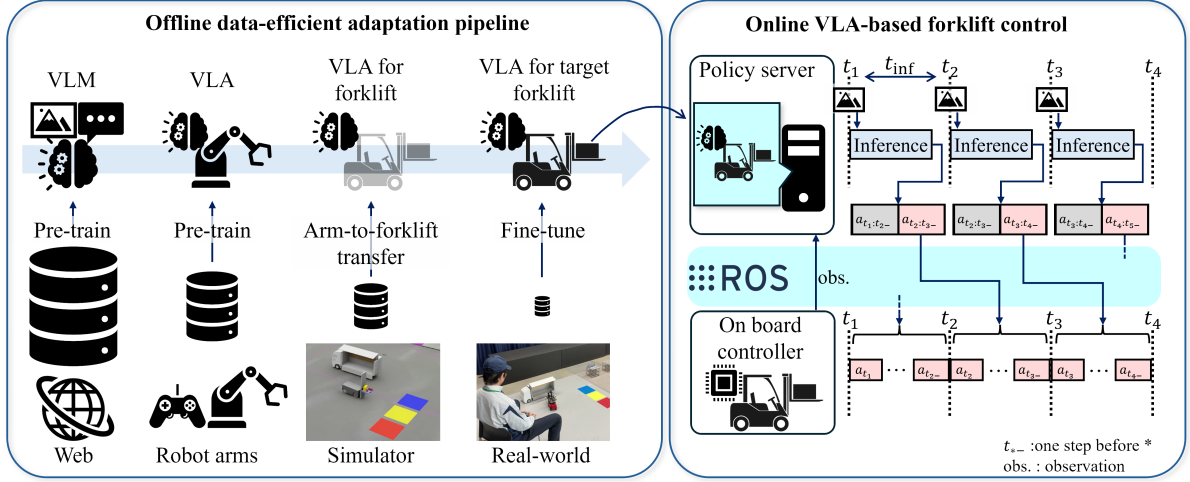


Fig. 2: Integrated system. A VLA policy is first constructed using readily available web and manipulation data, and then adapted to the target forklift using simulator-generated and limited real-world data. In the online operation, inference and forklift execution run asynchronously to reduce latency. The system maintains temporal consistency by executing future action chunks predicted by the VLA.

$o_t := [I_t^l, I_t^r, \ell_t, q_t]$, where I_t^l and I_t^r are the left and right camera images, respectively; ℓ_t is the prompt; and q_t is a vector of the vehicle’s speed, yaw rate, and fork height. Given o_t , the VLM branch classifies the current subtask, while the action-expert branch generates an action sequence $a_{t:t+H} := (a_t, \dots, a_{t+H-1})$ over a horizon H , where $a_t \in \mathbb{R}^3$ includes commands for the throttle, steering angle, and fork height. We added BERT-style subtask classification via a multilayer perceptron (MLP) [20]. We defined seven subtasks $c_t \in \{1, \dots, 7\}$ as described in Section III-A. Following the formulation of $\pi_{0.5}$, we express our model as $\pi_\theta(a_t, c_t | o_t) = \pi_\theta(a_t | o_t)\pi_\theta(c_t | o_t)$. The formulation shows that the subtask prediction c_t does not directly affect action generation. Instead, the two objectives are coupled via the loss function described in the next section.

B. Action and subtask prediction training

In this study, we initialized our model with the $\pi_{0.5}$ weights pretrained on robot data and then fine-tuned it for forklift control. We optimized the model using action and subtask classification losses. Subtask classification serves as an auxiliary loss, acting as a regularizer to encourage actions that are consistent with the inferred subtask.

Forklift data are not publicly available, and collecting real-world data is expensive. Therefore, we performed two-stage fine-tuning: we first fine-tuned the model on a large-scale simulation dataset and then performed a short additional fine-tuning stage on a small real-world dataset.

Given the expected limited diversity of the data, we applied randomization to q_t during training. Specifically, q_t was perturbed using multiplicative gain noise and additive offset noise sampled from a normal distribution. This state randomization was independent of the data-side randomization.

In this study, we used only two prompts: “Stop and wait.”

and “Pick up the pallet on the truck. Find the {color} zone and place the pallet there.” The operator switched between these prompts and specified the target color. To improve robustness to observation ambiguity, we jittered the timing of prompt injection during both training and inference. Our model uses single-shot observations without history. During the forklift operation, as the camera images can remain nearly unchanged while the vehicle repeatedly moves forward and backward, selecting appropriate actions using only the current observation is challenging. This issue is particularly pronounced during pallet pickup, when the cameras mainly show the interior of the truck with little visual change. Therefore, we split the task instruction into two sub-prompts. During training, we first provide “Pick up the pallet on the truck.” and switch to “Find the {color} zone and place the pallet there.” at the transition from the “pickup” subtask to the “backout” subtask. During inference, this switch is triggered automatically when the fork height exceeds a specified threshold, indicating that the pallet has been lifted off the truck’s floor.

V. DATASET

We reduced the data collection cost by generating large-scale simulation data using a simulator. Additionally, we collected a small amount of real-world data on the target platform via human teleoperation. We collected 9.38 h of simulation data and 0.8 h of real-world data; thus, real-world data accounted for approximately 8% of the total duration.

A. Simulation data

We built the simulation environment in Isaac Sim using CAD models [21]. For the background, we used a 3D-scanned model of the actual environment. We matched the relative configuration of the wing truck, pallet, drop-off location, and forklift initial pose to the real task setup. We

then randomly placed this task set within the background model and collected o_t and a_t . To increase data diversity, we randomized the lighting intensity, the colors of the pallet and the load, the load arrangement, and the camera pose [22].

During simulation, we controlled the forklift using a hard-coded subtask manager and non-learning-based methods that leveraged privileged information available in the simulator. We randomly selected the target color, generated the corresponding subtask sequence, and transitioned to the next subtask when a predefined success threshold was satisfied. Using this setup, we collected 9.38 h of simulation data.

B. Real-world data

We collected real-world data through joystick teleoperation with the operator directly observing the scene. The control period was 20 ms, matching the simulation, and all data were logged at this rate. In total, we collected 0.8 h of real-world data.

VI. REAL-WORLD EXPERIMENT

We performed two-stage fine-tuning using large-scale simulation data, followed by a small amount of real-world data. To assess this design, we compared it with two baselines: a model fine-tuned using only real-world data (“Only real”) and a model fine-tuned using only simulation data (“Only sim.”). To mitigate overfitting on the small real-world dataset, we used fewer fine-tuning steps for “Only real.” In addition, we prepared a variant of the proposed method without subtask prediction by disabling the auxiliary subtask loss to evaluate its effect.

A. Setting

Each model was evaluated in 18 real-world trials conducted under two different background conditions. For each background, we executed the unloading task three times for each of the three color zones under different load placements on the pallet. A trial was successful if the forklift unloaded the pallet without dropping the load and the center of the pallet was within the target zone. Each trial started with the prompt “Stop and wait.” After the operator switched the prompt to the task instruction, no further human input was provided.

B. Deployment

As described in Section III-C, we performed inference on a ground computer. The onboard computer applied the inferred actions to the forklift every 20 ms. The ground computer was equipped with an Intel® Xeon® Gold 6146 processor (3.20 GHz, 24 cores), two NVIDIA RTX A6000 GPUs, and 96 GB RAM. Inference required approximately 100 ms with this setup; therefore, we ran the policy every 200 ms ($t_{\text{inf}} = 0.2$). We trained the model to predict a sequence of 20-ms actions up to 400 ms ($H = 20$). Accordingly, we executed the predicted actions corresponding to 200–400 ms after the observation, maintaining temporal consistency between observation and control.

TABLE I: Success rate

Proposed	Proposed (w/o subtask)	Only real	Only sim.
100 %	39%	50 %	0 %

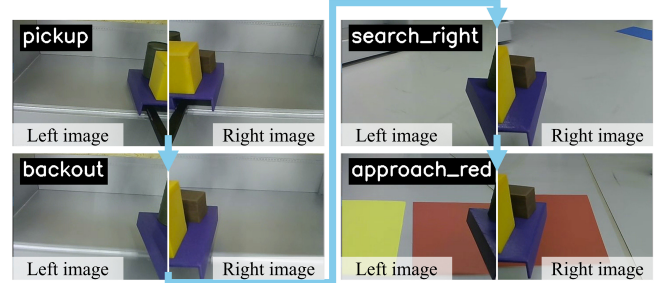


Fig. 3: Camera images in a real demonstration. White texts indicate the predicted results of the current subtask.

C. Results

Table I summarizes the task success rates. Here, “Proposed (w/o subtask)” denotes a variant trained without the auxiliary subtask loss. The proposed method succeeded in all 18 trials. In contrast, “Only real” achieved a 50% success rate, and “Only sim.” achieved 0%. Despite the very limited real-world data, “Only real” attained a 50% success rate, suggesting that the pretrained $\pi_{0.5}$ is a strong model. Additional training on large-scale simulation data in the proposed method enabled the policy to achieve a high success rate, even for forklift operation not covered by the original pretrained model. Fig. 3 shows example camera images obtained as the proposed model executes the task with subtask prediction. Fig. 1 shows a representative successful roll-out of the proposed model.

When subtask prediction was removed, our method achieved a lower success rate compared to that of “Only real.” The most frequent failure mode was transporting the pallet to the wrong color zone or to an unintended location. This suggests that without subtask prediction, the model fails to maintain consistency between the language goal and generated actions. This finding is consistent with that of prior work emphasizing the importance of VLM-side training [17].

The failure of “Only sim.” was likely due to insufficient data diversity, because the simulation data were generated using a single control policy.

VII. DISCUSSION AND CONCLUSION

This study presented a data-efficient integration of VLA control into a forklift unloading system. Motivated by the difficulty of automating sequential forklift operations with hard-coded logic and the lack of large-scale forklift operation data, we adapted a pre-trained VLA policy to forklift control using simulation-generated and limited real-world forklift data. The proposed integrated system connects VLA inference, phase prediction, asynchronous action-chunk execution, and low-level vehicle control. Through 72 real-world trials, we found that simulation-generated data supports policy adaptation, while subtask prediction helps align generated actions with task instructions.

However, the current evaluation remains limited in three respects. First, robustness to wider variations in environments, pallet poses, target layouts, or language instructions has not yet been established. Second, recovery behaviors and contact-rich manipulation required in practical logistics operations are not covered. Finally, the experiments were conducted on a 1/14-scale forklift platform; deployment on full-scale forklifts will require further validation under full vehicle dynamics, sensing noise, safety constraints, and data collection settings.

REFERENCES

- [1] M. Walter, S. Karaman, E. Frazzoli, and S. Teller, "Closed-loop pallet manipulation in unstructured environments," in *IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, 2010.
- [2] S. Teller *et al.*, "A voice-commandable robotic forklift working alongside humans in minimally-prepared outdoor environments," in *2010 IEEE Int. Conf. Robot. Autom. (ICRA)*, 2010, pp. 526–533.
- [3] S. Hadwiger, D. Kube, and T. Meisen, "Partial observability in vision-based forklift navigation with compressed visual information," *Int. J. Intell. Robot. Appl.*, pp. 1–24, 2026.
- [4] S. Yin *et al.*, "A survey on multimodal large language models," *Nat. Sci. Rev.*, vol. 11, no. 12, p. nwae403, 2024.
- [5] A. Brohan *et al.*, "RT-1: Robotics transformer for real-world control at scale," in *Robotics: Sci. Syst. (RSS)*, 2023, pp. 1–11.
- [6] B. Zitkovich *et al.*, "RT-2: Vision-language-action models transfer web knowledge to robotic control," in *Conf. Robot Learning (CoRL)*, vol. 229, 2023, pp. 2165–2183.
- [7] K. Kawaharazuka, J. Oh, J. Yamada, I. Posner, and Y. Zhu, "Vision-language-action models for robotics: A review towards real-world applications," *IEEE Access*, vol. 13, pp. 162 467–162 504, 2025.
- [8] M. J. Kim *et al.*, "OpenVLA: An open-source vision-language-action model," in *Conf. Robot Learning (CoRL)*, 2024.
- [9] A. O'Neill *et al.*, "Open x-embodiment: Robotic learning datasets and rt-x models : Open x-embodiment collaboration0," in *2024 IEEE Int. Conf. Robot. Autom. (ICRA)*, 2024, pp. 6892–6903.
- [10] K. Black *et al.*, " π_0 : A vision-language-action flow model for general robot control," *arXiv preprint arXiv:2410.24164*, 2024.
- [11] J. Bjorck *et al.*, "Gr00t n1: An open foundation model for generalist humanoid robots," *arXiv preprint arXiv:2503.14734*, 2025.
- [12] R. Iinuma, Y. Kojima, H. Onoyama, T. Fukao, S. Hattori, and Y. Nonogaki, "Pallet handling system with an autonomous forklift for outdoor fields," *J. Robot. Mechatron.*, vol. 32, no. 5, pp. 1071–1079, 2020.
- [13] J. Huemer *et al.*, "Adapt: An autonomous forklift for construction site operation," *arXiv preprint arXiv:2503.14331*, 2025.
- [14] S. Mae and H. Kataoka, "Efficient load interference detection with limited labeled data," in *IEEE/SICE Int. Symp. Syst. Integr. (SII)*, 2025, pp. 1422–1427.
- [15] K. Oishi, T. Kato, H. Makino, and S. Ito, "Visual-based forklift learning system enabling zero-shot sim2real without real-world data," in *2025 IEEE Int. Conf. Robot. Autom. (ICRA)*, 2025, pp. 4915–4921.
- [16] N. Karnchanachari *et al.*, "Towards learning-based planning: The nuplan benchmark for real-world autonomous driving," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*, 2024, pp. 629–636.
- [17] K. Black *et al.*, " $\pi_{0.5}$: a Vision-Language-Action model with open-world generalization," in *Conf. Robot Learning (CoRL)*, vol. 305, 2025, pp. 17–40.
- [18] K. Black, M. Galliker, and S. Levine, "Real-time execution of action chunking flow policies," *Advances in Neural Information Processing Systems*, vol. 38, pp. 33 383–33 407, 2026.
- [19] M. Quigley *et al.*, "Ros: an open-source robot operating system," in *ICRA workshop on open source software*, vol. 3. Kobe, Japan, 2009, p. 5.
- [20] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. NAACL-HLT*, 2019, pp. 4171–4186.
- [21] M. Mittal *et al.*, "Orbit: A unified simulation framework for interactive robot learning environments," *IEEE Robot. Autom. Lett.*, vol. 8, pp. 3740–3747, 2023.
- [22] J. Tobin, R. Fong, A. Ray, J. Schneider, W. Zaremba, and P. Abbeel, "Domain randomization for transferring deep neural networks from simulation to the real world," in *IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, 2017, pp. 23–30.